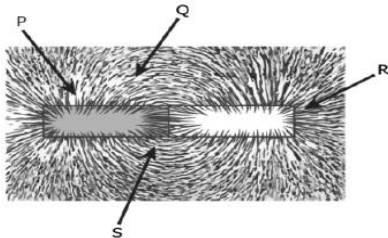


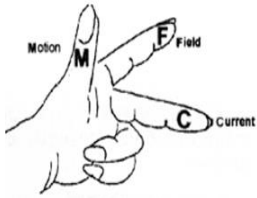
Kendriya Vidyalaya Sangathan, Varanasi Region Pre-Board-I Examination-2024-25		
Class-X		Max. Marks-80
Subject-Science (086)		Max. Time- 3 Hrs.

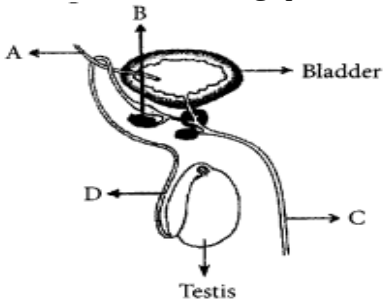
General Instructions:

- This question paper consists of 39 questions in 5 sections.*
- All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of these questions.*
- Section A consists of 20 objective type questions carrying 1 mark each.*
- Section B consists of 6 Very Short questions carrying 02 marks each. Answers to these questions should be in the range of 30 to 50 words.*
- Section C consists of 7 Short Answer type questions carrying 03 marks each. Answers to these questions should be in the range of 50 to 80 words.*
- Section D consists of 03 Long Answer type questions carrying 05 marks each. Answer to these questions should be in the range of 80 to 120 words.*
- Section E consists of 3 source-based/case-based units of assessment of 04 marks each With sub-parts.*

Section-A		
Question 1 to 16 are multiple choice questions. Only one of the choice is correct. Select and write the correct choice as well as the answer to these questions.		
Q. Nos.	Questions	Marks
1	In a redox reaction, oxidation refers to: a) Gain of electrons b) Loss of oxygen c) Loss of electrons d) Addition of hydrogen	1
2	Which of the following substances is a base? a) Sodium chloride (NaCl) b) Potassium hydroxide (KOH) c) Carbon dioxide (CO ₂) d) Water (H ₂ O)	1
3	Which of the following salts does not have water of crystallization? a) Blue vitriol b) Baking soda c) Washing soda d) Gypsum	1
4	The difference of water molecules in gypsum and Plaster of Paris is a) 5/2 b) 2 c) 1/2 d) 3/2	1
5	An iron nail was suspended in CuSO ₄ solution and kept for a while the solution is a) Remained blue and coating was found on the nail. b) Turned green and a coating was formed on the nail c) Remained blue and no coating was formed on the nail d) Turned green and no coating was formed on the nail	1
6	The IUPAC name of the compound CH ₃ -CH ₂ -OH is a) Methanol b) Ethanol c) Propanol d) Butanol	1

16	Shreya places some iron filings around a magnet. The iron fillings arrange themselves as shown in the image below:  Shreya labelled four different regions around the magnet. Where would the magnetic field be the strongest? a) P b) Q c) R d) S	1
QuestionNo.17 to 20 consists of two statements–Assertion (A) and Reason(R). Answer these questions by selecting the appropriate option given below: A. Both A and R are true, and R is the correct explanation of A. B. Both A and R are true, and R is not the correct explanation of A. C. A is true but R is false. D. A is false but R is true		
17	Assertion (A): The direction of the magnetic field lines around a straight conductor can be determined using the Right-Hand Thumb Rule. Reason (R): The Right-Hand Thumb Rule states that the thumb points in the direction of the magnetic field.	1
18	Assertion (A): A current carrying wire deflects a magnetic needle placed near it. Reason (R): A magnetic field exists around a current carrying wire.	1
19	Assertion (A): The concentration of harmful chemicals is least in human beings. Reason (R): Man is at the apex of the food chain.	1
20	Assertion (A): Deforestation contributes to an increase in carbon dioxide levels in the atmosphere. Reason (R): Trees absorb carbon dioxide during photosynthesis.	1
Section-B QuestionNo.21 to 26 are very short answer questions		
21	State the difference between exothermic and endothermic reactions with an example for each. OR What is a precipitation reaction? Give an example.	2
22	What are amphoteric oxides? Give examples?	2
23	What is cellular respiration? During cellular respiration, the complete oxidation of one molecule of glucose ($C_6H_{12}O_6$) releases approximately 38 ATP molecules. If a cell metabolizes 5 molecules of glucose, how many	2

	ATP molecules are produced? OR In a certain experiment, a scientist measured the rate of transpiration in a plant and found that it was 3 ml. per hour. If the rate remains constant, how much water will be lost through transpiration over a week? Write the source of raw materials required by the plants for photosynthesis?	
24	If a pure tall pea plant (TT) is crossed with a pure short pea plant (tt), what will be the phenotypic and genotypic ratio of the F1 generation?	2
25	Stars appear to twinkle, what is the scientific reason? Why don't planets twinkle like stars?	2
26	If the length of a wire is doubled, how does its resistance change, assuming the material and cross-sectional area remain the same?	2
Section-C		
Question No. 27 to 33 are short answer questions		
27	You have two solutions 'A' and 'B'. The pH of solution 'A' is 6 and pH of solution 'B' is 8. Which solution has more hydrogen ions concentration? Which is acidic and which one is basic?	3
28	Explain why calcium metal after reacting with water starts floating on its surface. Write the chemical equation for the reaction. Name one more metal that starts floating after some time when immersed in water.	3
29	Draw structure of a neuron and write the function of its main parts. OR What are the major parts of the brain and write their respective functions.	3
30	Distinguish between genotype and phenotype. Write an example of each.	3
31	Two students of the class see nearby objects as blurred. What is the reason behind it and how it can be corrected? Explain with diagram	3
32	A student is unable to determine the direction of magnetic field produced around a straight conductor carrying current. How can you help him and state the law behind it OR One rule is depicted in the figure given below. Identify this rule, for what  this rule is responsible and state this rule.	3
33	a) In a food chain, if 25,000 joules of energy is available to the producer, how much energy will be available to the secondary consumer to transfer it to the tertiary consumer? b) Natural water bodies are not regularly cleaned whereas an aquarium	1x3

	needs regularly cleaning. Why? c) Why the number of trophic levels in a food chain is limited	
Section-D		
Question No.34 to 36 are long answers questions.		
34	a) What are covalent bonds? Explain with an example. b) Why are covalent compounds generally poor conductors of electricity? c) Explain the difference between a single, double, and triple covalent bond with examples. d) Write the electron dot structure of methane (CH₄). OR a) Define homologous series? Explain with an example. b) Explain the versatile nature of carbon with reference to catenation and tetravalency. c) Write the chemical equations for the combustion of ethane. d) Differentiate between saturated and unsaturated hydrocarbons.	1+1+2 +1=5
35	a) Name the male and female reproductive parts of a flower. b) Describe the process of pollination. c) What do you mean by fertilization in plants? d) How does the formation of seeds take place after fertilization? OR Answer the following questions which is based on the diagram given below:  a) Name the C and D parts as shown in above figure b) Name the hormone secreted by testis and mention its role. c) State the functions of part B d) What is the role of part C in process of reproduction	1+2 +1+1= 5
36	Draw ray diagrams showing the image formation by a convex mirror when an object is placed (a) at infinity (b) at finite distance from the mirror. Calculate the distance at which a convex lens of focal length 20 cm must be placed from an object to form a real image at a distance of 30 cm from the lens. OR	2+3=5

	List two characteristics of the images formed by plane mirrors. An object is placed at a distance of 10 cm from a convex mirror of focal length 15 cm. Find the position and nature of the image.	
Section –E Question No.37 to 39 are case-based/data –based questions.		
37	Pratiksha heats a sample of calcium carbonate in a test tube. After heating, a gas is evolved that turns lime water milky. The residue left in the test tube is calcium oxide. <u>Attempt either sub part B or C.</u> A. Name the gas released during the heating of calcium carbonate. (1) B. Write the balanced chemical equation for the decomposition of calcium carbonate. (2) OR C. What do you mean by decomposition reaction? D. Why does the gas turn lime water milky? (1)	4
38	Avnish conducts an experiment where two identical potted plants are placed in sunlight. One plant is covered with a black cloth for 48 hours, while the other remains uncovered. <u>Attempt either sub part A or B.</u> A. What is the purpose of covering one plant with black cloth? (2) OR B. What do you predict will happen to the covered plant after 48 hours? C. What is the role of chlorophyll in the process of photosynthesis. (1) D. How does this experiment demonstrate the necessity of light for Photosynthesis? (1)	4
39	A student connects two resistors, 4 Ω and 6 Ω, in series with a 24 V battery and wants to analyze the circuit. A. What is the equivalent resistance of the circuit? (1) B. What is the current flowing through the circuit? (1) <u>Attempt either sub part C or D.</u> C. What is the potential difference across the 4 Ω resistor? (2) OR D. If the resistors were connected in parallel, what would be the total current?	4